*A Project Based Learning (PBL) Report*

***on***

# "LPG Gas Leakage Detection with Auto – Regulator On / Off System & Temperature and Humidity Sensing"

By


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Guide

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Sinhgad Institutes

# *Certificate*

***This is to certify that the PBL report entitled***

## "LPG Gas Leakage Detection with Auto – Regulator On / Off System & Temperature and Humidity Sensing"

of

TE *(Electronics & Telecommunication)*

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Date:

# *Contents*

# *1. Abstract*

LPG leakages are a mutual hindrance in household and manufacturing nowadays. It is very life threatening if you will not distinguish and modified right away. The idea behind our project is to give a solution by power cut the gas provision as soon as a gas leakage is perceived apart from activating the sounding alarm. With this we had a feature of displaying Environmental Temperature & Humidity.

# *2. Introduction*

The project entitled "LPG Gas Leakage Detection with Auto – Regulator On / Off System & Temperature and Humidity Sensing", will be a great help in terms of preventing any danger caused by gas leakage. The purpose of this project is to detect the presence of LPG leakage as a part of a safety system. Apart from sound alarm, the solenoid valve will be triggered to shut down the gas supply to prevent any harmful effects due to gas leakage. Descriptively, we use a gas sensor to monitor the LPG if the gas leak reaches beyond the normal level. This proposed project will trigger the sound alarm. The people can be saved from a potential explosion caused by gas leakage.

# *3. Specifications*

## **Arduino UNO (ATmega328P) :-**

| | |
|---|---|
| Microcontroller | ATmega328P – 8 bit AVR family microcontroller |
| Operating Voltage | 5V |
| Recommended Input Voltage | 7-12V |
| Input Voltage Limits | 6-20V |
| Analog Input Pins | 6 (A0 – A5) |
| Digital I/O Pins | 14 (Out of which 6 provide PWM output) |
| DC Current on I/O Pins | 40 mA |
| DC Current on 3.3V Pin | 50 mA |
| Flash Memory | 32 KB (0.5 KB is used for Bootloader) |
| SRAM | 2 KB |
| EEPROM | 1 KB |
| Frequency (Clock Speed) | 16 MHz |

## **Sensors Specifications :-**

### 1) **MQ6 ( Gas Sensor ) :-**

- Operating Voltage is +5V.
- Can be used to detect LPG or Butane Gas.
- Analog output voltage = 0-5V.
- Digital output voltage = 0-5V (TTL Logic).
- Preheat Duration = 20 seconds.
- Can be used as a Digital or Analog sensor.
- The sensitivity of Digital pin can be varied using Potentiometer.

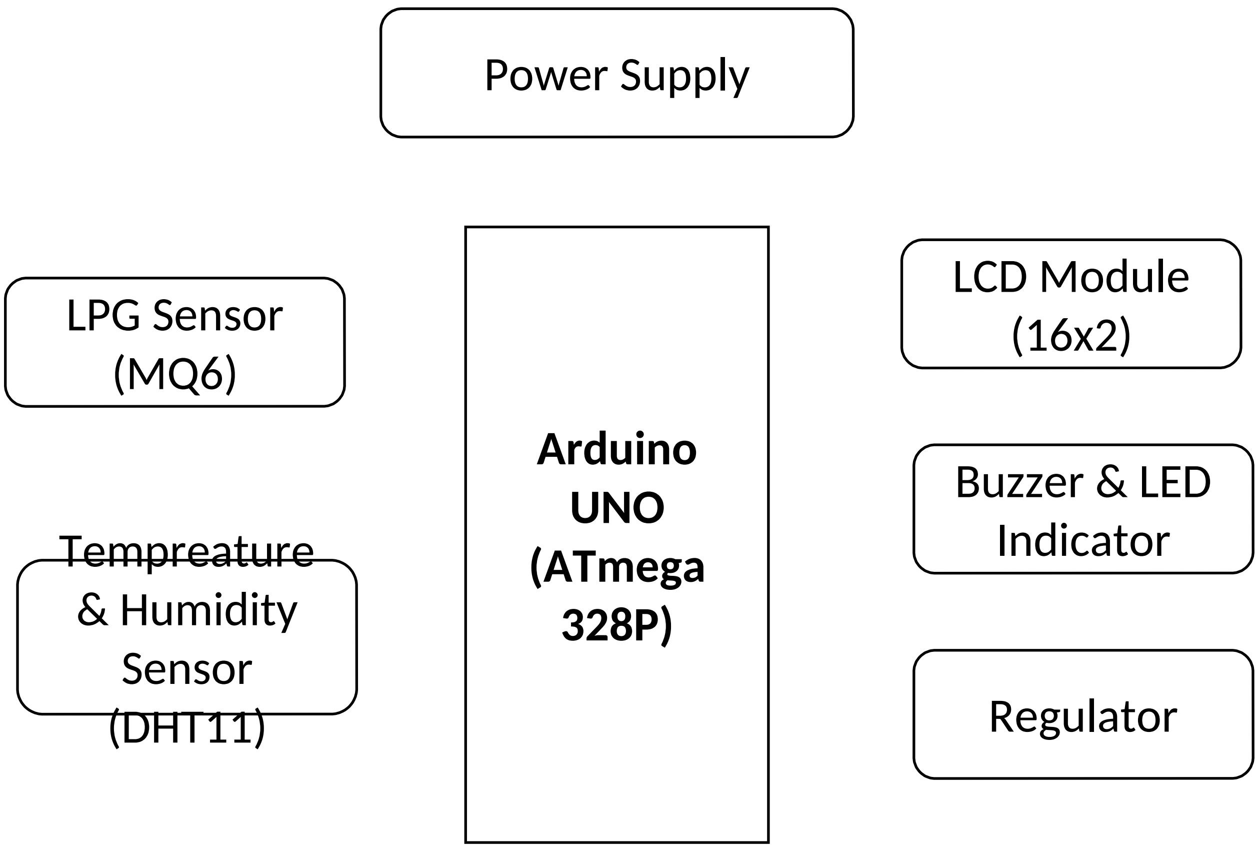
### 2) **DHT11 (Temperature & Humidity Sensor) :-**

- Operating Voltage: 3.5V to 5.5V
- Operating current: 0.3mA (measuring) 60uA (standby)
- Output: Serial data
- Temperature Range: 0°C to 50°C
- Humidity Range: 20% to 90%
- Resolution: Temperature and Humidity both are 16-bit
- Accuracy: ±1°C and ±1%

### 3) **16x2 LCD :-**

- Operating Voltage is 4.7V to 5.3V
- Current consumption is 1mA without backlight
- Alphanumeric LCD display module, meaning can display alphabets and numbers
- Consists of two rows and each row can print 16 characters.
- Each character is build by a 5×8 pixel box
- Can work on both 8-bit and 4-bit mode
- It can also display any custom generated characters
- Available in Green and Blue Backlight

# *4. Block Diagram and Description*



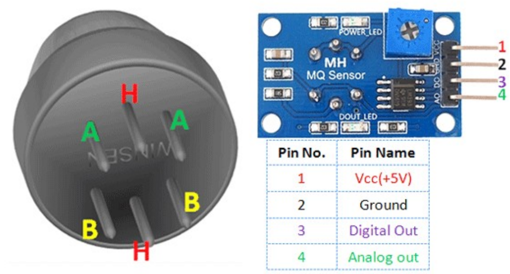

**Power Supply :-**

This block gives the input power to the controller & all other blocks of the project. The input supply between 0V – 12 V is given to the blocks as required.

**LPG Gas Sensor (MQ-6):-**

The MQ-6 Gas sensor can detect or measure gases like LPG and butane. In this Project we've used to detect LPG gas. The MQ-6 sensor module comes with a Digital Pin which makes this sensor to operate even without a microcontroller and that comes in handy when you are only trying to detect one particular gas. Using a MQ sensor it detection of gas is very easy. In this project, as the leakage is detected by the sensor, the signal is sent to the controller &then the controller responds to that signal.

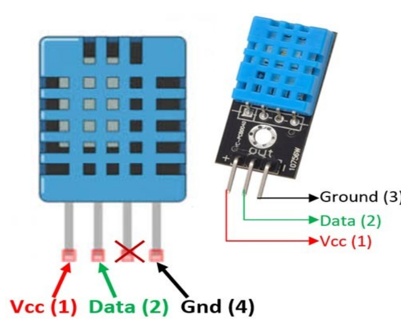
**MQ-2 Gas Sensor**




**Pin out of MQ-2 Gas Sensor**

**Temperature & Humidity Sensor (DHT-11) :-**

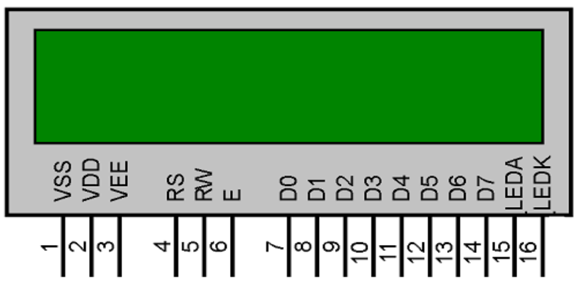
The DHT11 is a commonly used Temperature and humidity sensor. The sensor comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor is also factory calibrated and hence easy to interface with other microcontrollers. The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of ±1°C and ±1%. As DHT11 Sensor is factory calibrated and outputs serial data and hence it is highly easy to set it up.



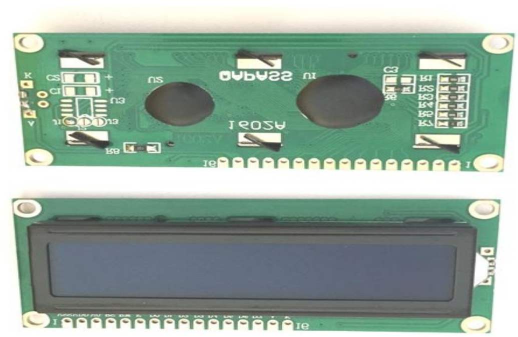

**Pin out of DHT-11**

**16x2 LCD :-**

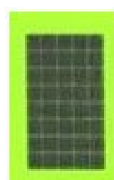
We come across LCD displays everywhere around us. Computers, calculators, television sets, mobile phones, digital watches use some kind of display to display the time. An LCD is an electronic display module which uses liquid crystal to produce a visible image. The 16×2 LCD display is a very basic module commonly used in DIYs and circuits. The 16×2 translates o a display 16 characters per line in 2 such lines. In this LCD each character is displayed in a 5×7 pixel matrix.




**Pin out of 16x2 LCD**



**16x2 LCD Module**



**5x7 Pixel Matrix**

**Buzzer & LED Indicator :-**

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric (piezo for short). Typical uses of buzzers and beepers include alarm devices, timers, and confirmation of user input such as a mouse click or keystroke. In this project, Buzzer is used to give the indication of gas leakage by sounding an alarm. As the gas leakage is detected by the MQ-2 sensor, the controller will give the control signal to the buzzer & LED indicator. Due to this the Buzzer will give the sound alarm with the glow of LED.

## Buzzer

## LED's

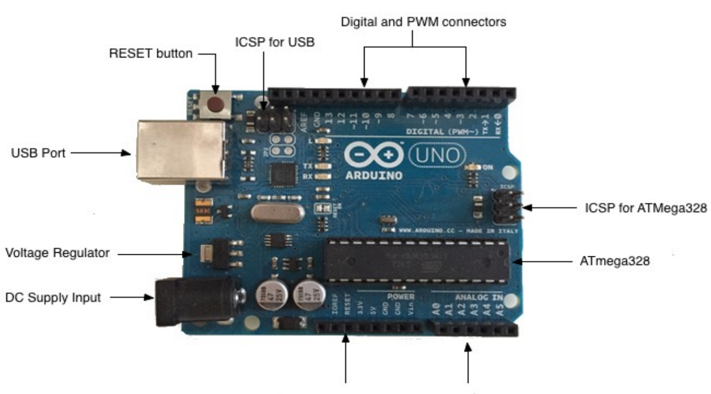
**Regulator (Relay) :-**

A relay is an electrically operated switch. It consists of a set of input terminals for a single or multiple control signals, and a set of operating contact terminals. Relays are used where it is necessary to control a circuit by an independent low-power signal, or where several circuits must be controlled by one signal. In this project we've connected a relay instead of solenoid valve. This relay is used to cut-off the Gas supply as the leakage is detected. As the sensor senses the LPG gas leakage, the controller will give the signal to relay due to which the Gas supply from the cylinder will be disconnected. Relay is having various types i.e with different channels such as 1-channel relay, 2-channel relay, 4-channel relay, etc. In this project we are using 1-channel relay, as we are having only one regulator connected to cylinder. Single Channel Relay Module is as shown below,

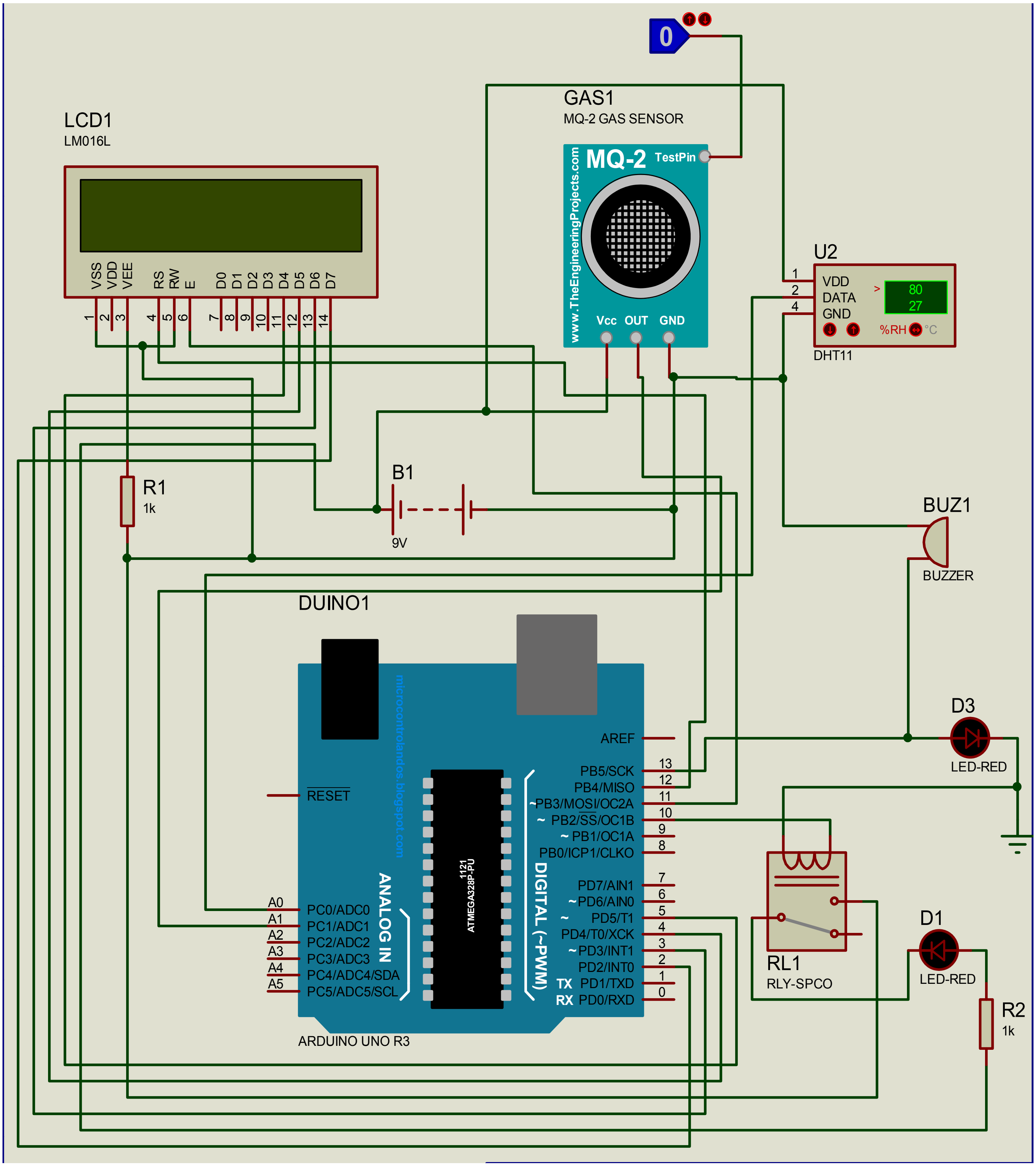

**Single channel relay Module**

**Controller [ Arduino UNO (ATmega 328P)] :-**

Arduino UNO is a controller board with a programming IC (ATmega 328P) mounted on it. This controller is programmed in a certain language that the controller understands. The function we've to perform by this is done by programming a controller with certain instructions. This controller consists of various digital I/O pins, Analog input pins (to connect the sensors), power supply pins, USB port, Voltage regulator, reset switch, Connectors, & mainly a programming IC ATmega328P. The Arduino UNO board & ATmega328P is as shown below,




# 5. Interfacing Diagram



# 6. *Software design / Program*

```
#include <LiquidCrystal.h>

#include <dht.h>


LiquidCrystal lcd (12, 11, 5, 4, 3, 2);


#define lpg_sensor A1

#define ledPin 13

#define reLay 10

#define dht_apin A0                    // Analog Pin sensor is connected to

dht DHT;


void setup ()

 {

  Serial.begin(9600);

  delay(100);                          //Delay to let system boot


  pinMode(lpg_sensor, INPUT);

  pinMode(ledPin, OUTPUT);

  pinMode(reLay,OUTPUT);

  lcd.begin(16, 2);

  lcd.clear();

  lcd.setCursor(0,0);

  lcd.print("LPG Gas Detector");

   lcd.setCursor(1,1);
```

```
  lcd.print("E&TC  TE – C4");

  delay(5000);

  lcd.clear();

}


void loop()

 {

  DHT.read11(dht_apin);


    delay(50);                                   //Wait 5 seconds before accessing sensor again.


    if(digitalRead(lpg_sensor))

   {

    digitalWrite(ledPin, HIGH);

    digitalWrite(reLay, LOW);

    lcd.clear();

    lcd.setCursor(0,0);

    lcd.print("LPG Gas Leakage");

    lcd.setCursor(4, 1);

    lcd.print("     Alert     ");

    delay(500);

   }


  else

   {

    digitalWrite(ledPin, LOW);

    digitalWrite(reLay, HIGH);

    lcd.clear();
```

```
    lcd.setCursor(2,0);

    lcd.print("No Leakage ");

    delay(3000);


    lcd.clear();

    lcd.setCursor(1,0);

    lcd.print("Temp = ");

    lcd.print(DHT.temperature);          //printing temperarture to the LCD display

    lcd.print("'C");


    lcd.setCursor(0,1);

    lcd.print("Humi. = ");

    lcd.print(DHT.humidity);             //printing humidity to the LCD display

    lcd.print(" %");

    delay(3000);

  }


}
```

# 7. Result

### When no leakage detected :-

In this project, if there is no leakage detected by the sensor, the controller will read the value of the Temperature & Humidity Sensor (DHT-11) & Display the value of the temperature & humidity in the surrounding environment on LCD display & with that the message of " No Leakage " is displayed on LCD display with some delay, & this continuous in loop until the leakage is detected. The working diagram for this is shown below,

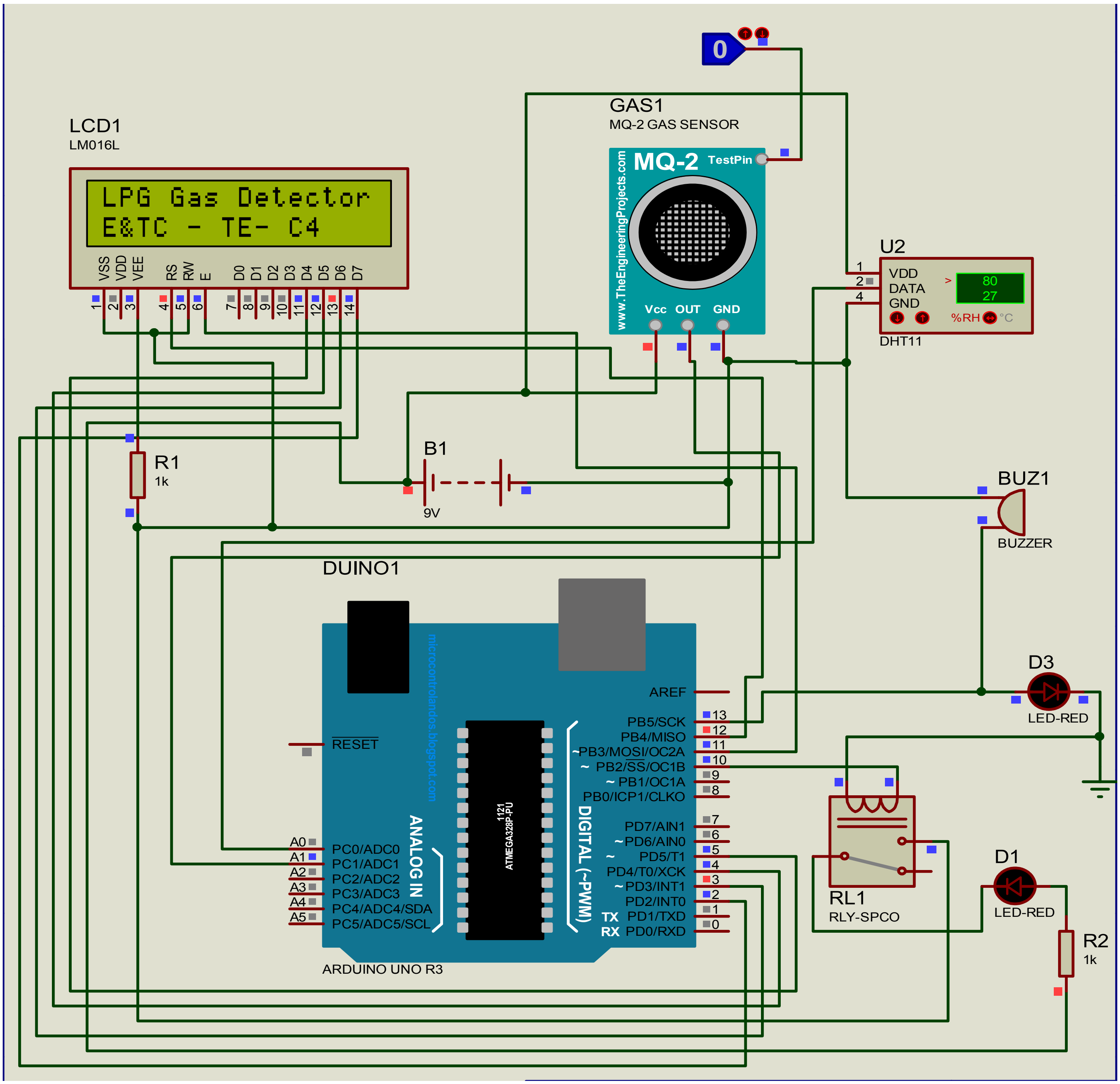
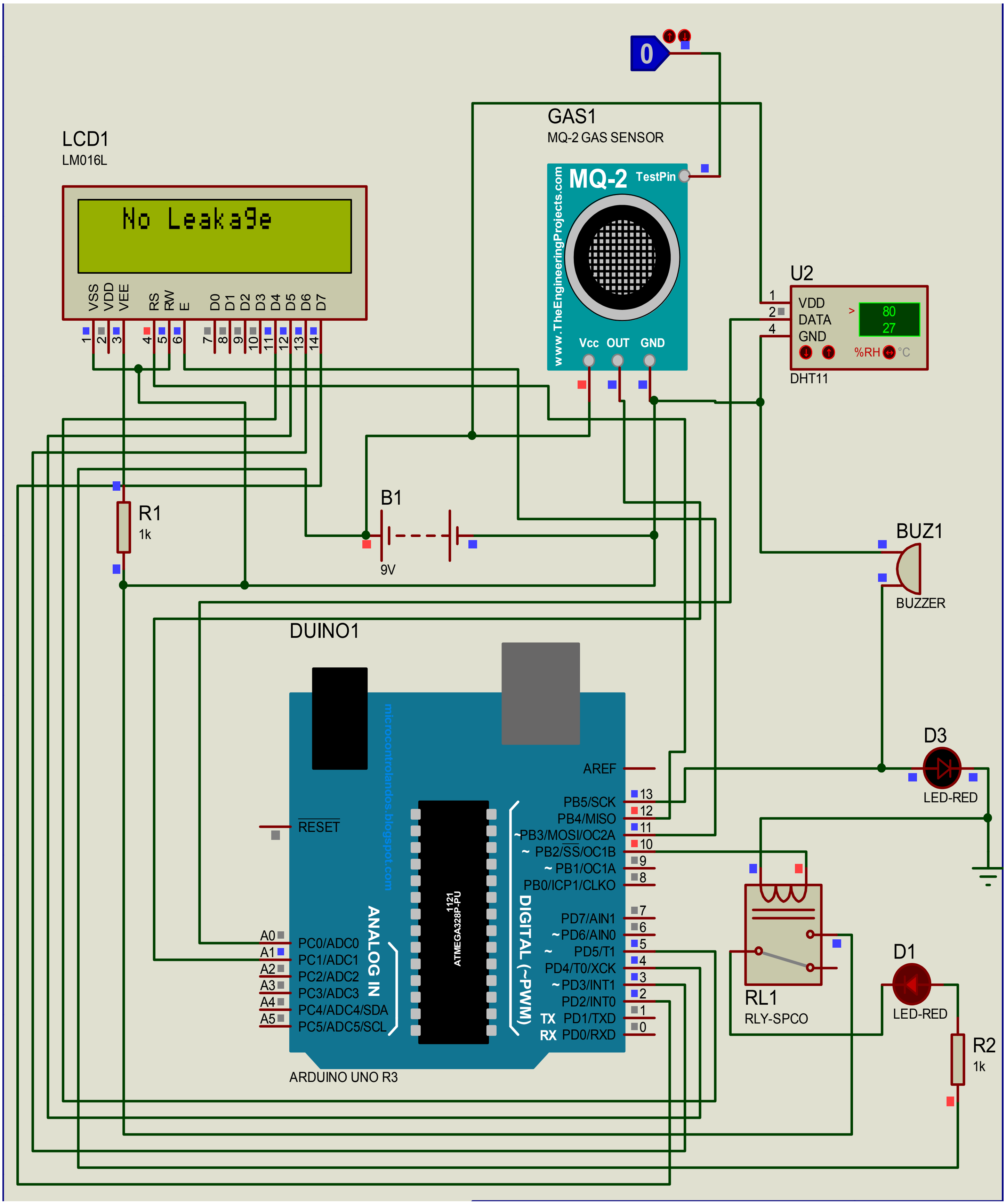
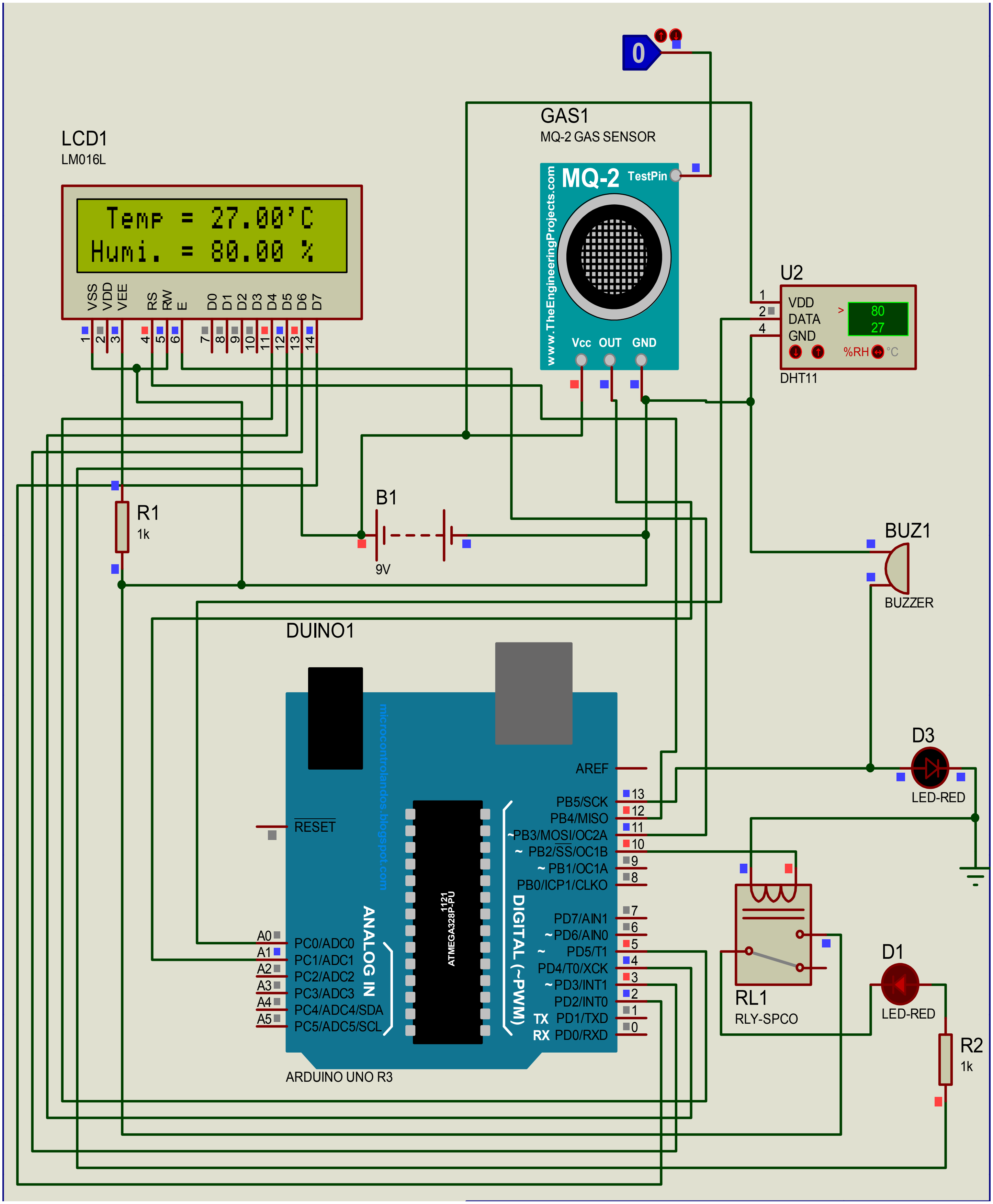



**fig. - Output when the project is switched ON**

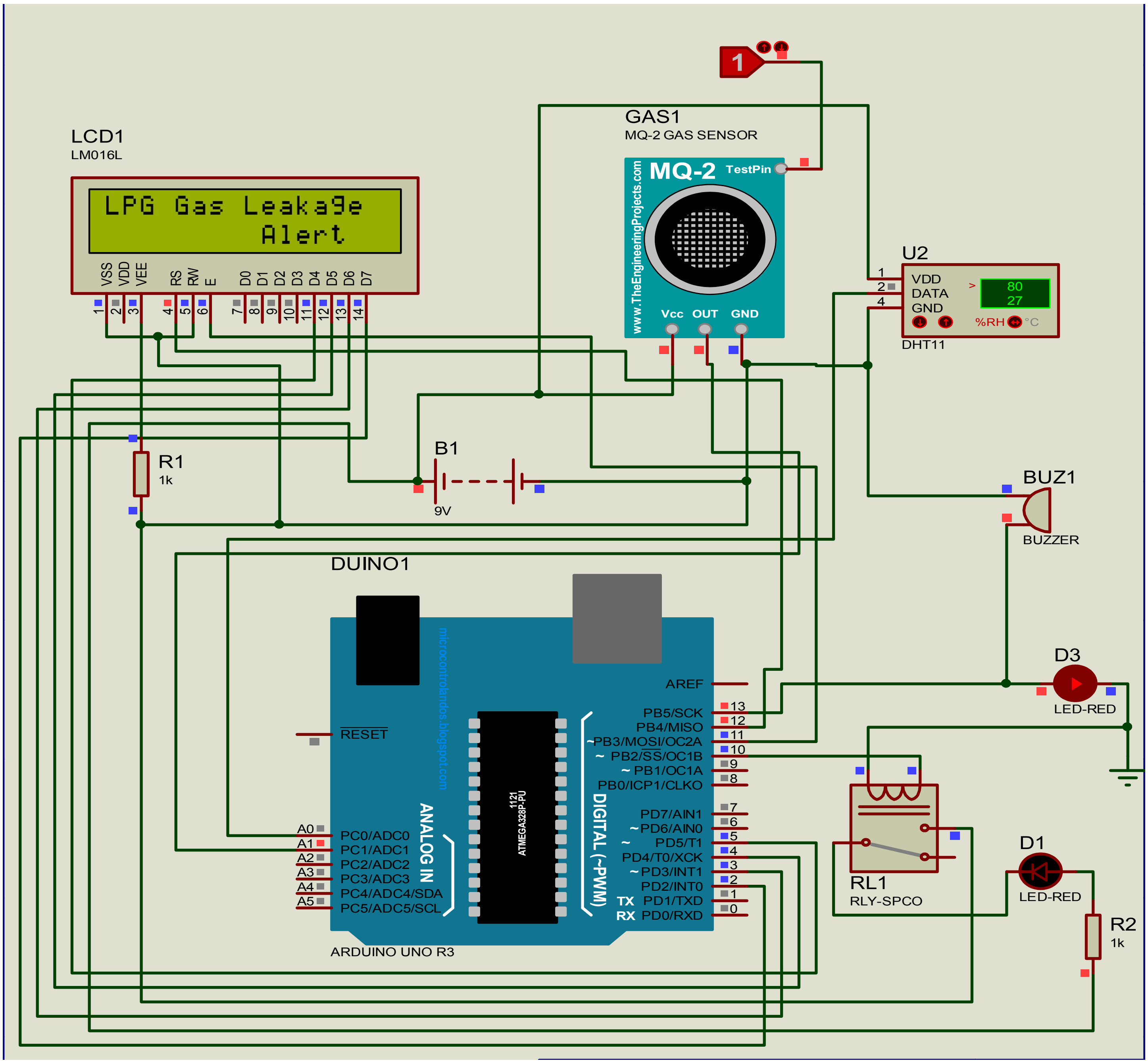

LCD1
LM016L
No Leakage
VSS VDD VEE RS RW E D0 D1 D2 D3 D4 D5 D6 D7
GAS1
MQ-2 GAS SENSOR
MQ-2 TestPin
www.TheEngineeringProjects.com
Vcc OUT GND
U2
VDD
DATA
GND
%RH
°C
DHT11
80
27
R1
1k
B1
9V
BUZ1
BUZZER
DUINO1
AREF
RESET
PB5/SCK
PB4/MISO
~PB3/MOSI/OC2A
~ PB2/SS/OC1B
~ PB1/OC1A
PB0/ICP1/CLKO
PD7/AIN1
~PD6/AIN0
~ PD5/T1
PD4/T0/XCK
~PD3/INT1
PD2/INT0
TX PD1/TXD
RX PD0/RXD
A0 PC0/ADC0
A1 PC1/ADC1
A2 PC2/ADC2
A3 PC3/ADC3
A4 PC4/ADC4/SDA
A5 PC5/ADC5/SCL
ANALOG IN
DIGITAL (~PWM)
ATMEGA328P-PU
1121
microcontrolandos.blogspot.com
ARDUINO UNO R3
D3
LED-RED
RL1
RLY-SPCO
D1
LED-RED
R2
1k




**When Leakage is Detected :-**

Now, if the gas leakage is detected by the MQ-2 gas sensor, the controller will send the control signal to the buzzer & LED (D3). This will turn on the Buzzer  & LED (D3), due to which the Buzzer gives the indication of Gas leakage by giving a big alarm sound & RED-LED (D3) gets turned on indicating the leakage.

As the buzzer & LED (D3) gets turned ON, the controller will also give the control signal to the relay (Solenoid), which will turn off the regulator / valve & the gas supply to the particular appliance will be get disconnected. In this project we've connected a RED-LED (D1) in place of regulator / valve to reduce the cost of the project. After turning OFF the regulator, the message " LPG Gas Leakage Alert " will be displayed on the LCD screen. The working diagram for this is shown below,



# *8. Advantages and Applications*

**Advantages :-**

- It is used in house as LPG leakage detection.
- It also detects alcohol so it is used as liquor tester.
- The sensor has excellent sensitivity combined with a quick fast response time.
- The system is highly reliable, tamper-proof and secure.
- In the long run the maintenance cost is very less when Compared to the present systems.
- It is possible to get instantaneous results and with high accuracy.

**Applications :-**

- Protection from any gas leakage in cars.
- For safety from gas leakage in heating gas fired appliances like boilers, domestic water heaters.
- Large industries which uses gas as their production.
- For safety from gas leakage in cooking gas fired appliances like ovens, stoves etc.

# *9. Conclusions & future Scope*

**<u>Conclusion</u> :-**

Gas leakage leads to severe accidents resulting in material losses and human injuries. Gas leakage occurs mainly due to poor maintenance of equipments and inadequate awareness of the people. Hence, LPG leakage detection is essential to prevent accidents and to save human lives. This system triggers LED and buzzer to alert people when LPG leakage is detected. This system is very simple yet reliable.

**<u>Future Scope</u> :-**

With recent development in technology, very interesting and significant improvement would be to accommodate multiple receiver MODEMS at different positions in the geographical area carrying duplicate SIM cards. Multilingual display can be another added variation in the project. Audio output can be introduced to make it user Friendly.